

KubeQuest - User Guide

Welcome to your Kubernetes learning environment! You have been allocated a dedicated cluster of 3 EC2 instances running on AWS with an Amazon Linux image. Your environment is isolated, secure, and ready for you to build on.

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Region per group

Each group has an assigned region for its infrastructure. Make sure to use the correct region for your group in AWS Console and CLI.

Group 01 to 12: eu-west-3 (Paris)

Group 13 to 24: eu-west-1 (Ireland)

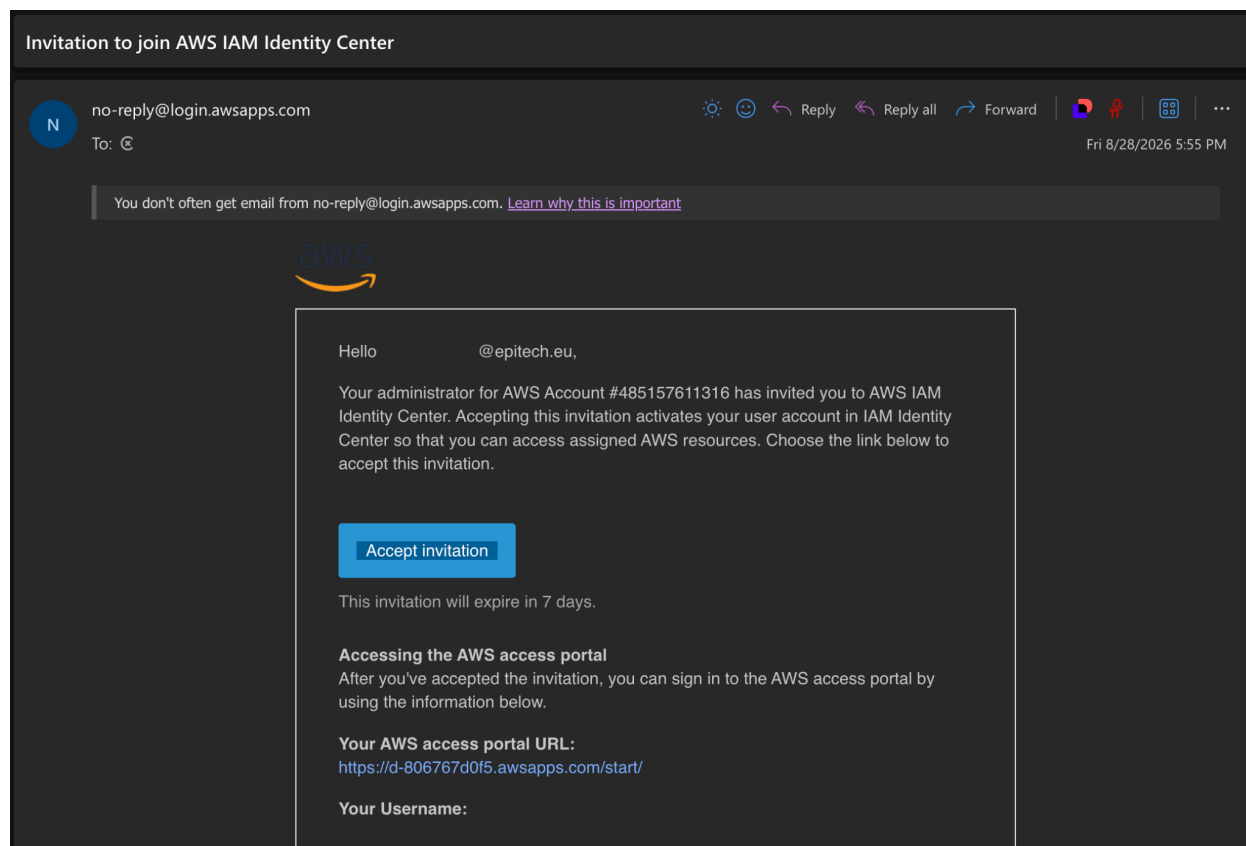
Group 25 to 36: eu-central-1 (Frankfurt)

Group 37 to 48: eu-west-2 (London)

Group 49 to 60: eu-north-1 (Stockholm)

Step 1: Setup your account

You should have received the following e-mail to activate your account in Epitech's AWS Organization:



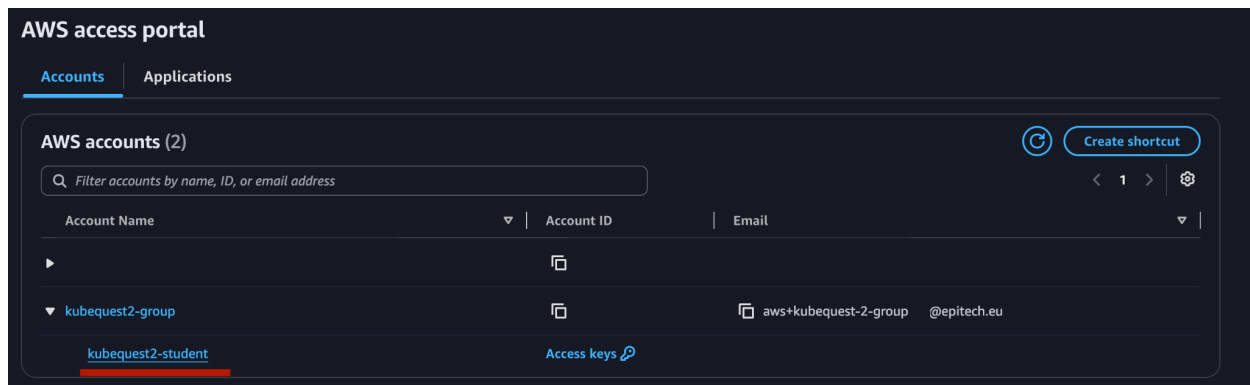
You will be asked for your username (your epitech email address), a password and to set up an MFA.

Step 2: Connect to the console

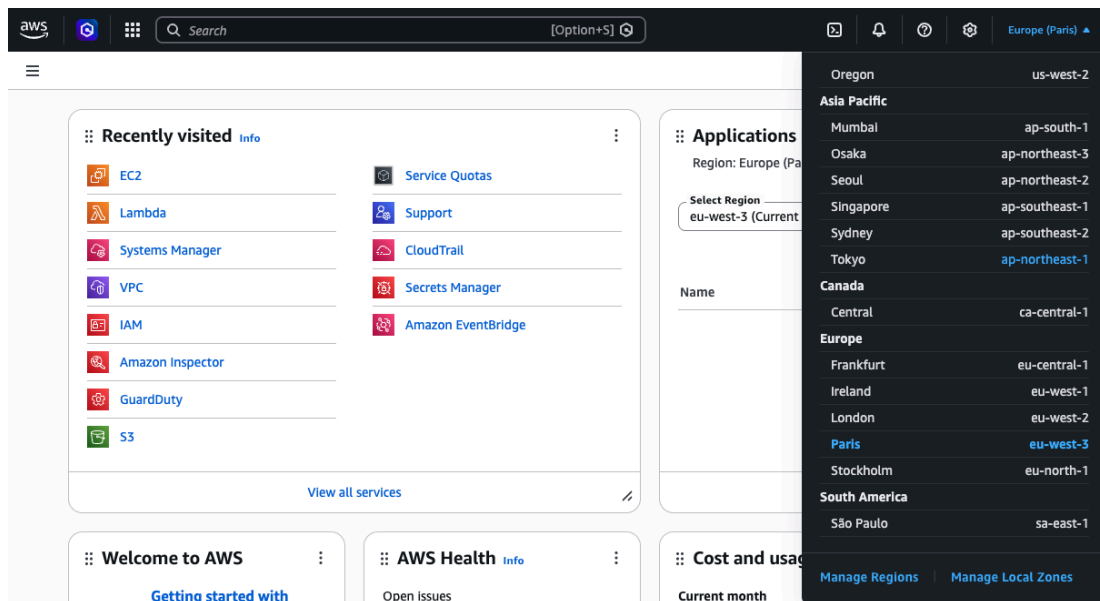
AWS access portal URL: <https://d-806767d0f5.awsapps.com/start/>

Login into Epitech's AWS access portal with your previously created account.

You should then have a page where you see the AWS accounts you have access to, you can expand and click on the role `kubequest2-student` to open the console of this account.



Once logged in, look at the top-right corner of the AWS Console and make sure you are in your assigned region.



Step 3: Setup your CLI

Follow these instructions if you want to interact with AWS through the CLI.

Install AWS CLI v2 for your operating system:

<https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>

Once installed, you can run this command that will open your browser to complete the setup:

...

```
$ aws configure sso
```

```
SSO start URL [None]: https://ssoins-66566efc3ab8cb5f.portal.eu-west-3.app.aws
```

```
SSO region [None]: eu-west-3
```

Attempting to automatically open the SSO authorization page in your default browser.

If the browser does not open, open the following URL: <link>

[select your AWS kubequest account]

The only role available to you is: kubequest2-student

Using the role name "kubequest2-student"

```
Default client Region [eu-xxxx-x]: <use the region you were assigned to>
```

```
CLI default output format (json if not specified) [None]:
```

```
Profile name [kubequest2-student-xxx]: kubequest2
```

...

Now you should be able to run this command to authenticate each day:

...

```
$ aws sso login --profile kubequest2
```

...

And then you should be able to run any aws cli commands using `--profile kubequest2` like this:

...

```
$ aws --profile kubequest2 ec2 describe-instances
```

...

Note: If you don't get any instances in the list, you probably misconfigured the default region.

Step 4: Start / Stop your instances

1. In the AWS Console, navigate Inside **EC2 > Instances**
2. Select the instance and click on Instance state:

Instances (1/80) Info Last updated 39 minutes ago Connect Instance state Actions Launch Instances

Find Instance by attribute or tag (case-sensitive)

group-01 Clear filters

Name	Instance ID	Instance state	Instance type	Alarm status	Availability zone
☒ ec2-group-01-node-1	i-023c81a1c1fc091ad	Running	t4g.medium	View alarms +	eu-west-3a
☐ ec2-group-01-node-2	i-04646d59e796c29cd	Stopped	t4g.medium	View alarms +	eu-west-3a
☐ ec2-group-01-node-3	i-068bb1c8fade5c79c	Stopped	t4g.medium	View alarms +	eu-west-3a
☐ ec2-group-01-node-4	i-039763472d3886fb3	Stopped	t4g.medium	View alarms +	eu-west-3a

Stop Instance
Start Instance
Reboot instance
Hibernate Instance
Terminate (delete) instance

Note: You only have permission to start, shutdown and reboot the instance. You don't have permission to terminate instances.

Step 5: Connecting to your instance

Your group has 3 instances. The first one (**ec2-group-XX-node-1**) has a public Elastic IP attached to it, this means that the IP will persist after the shutdown/restart of the instance. The two other instances have dynamic IPs which change after each restart.

There are two ways for accessing the instances: Session Manager (direct in the web UI).

Accessing using the Console

1. Select the instance that you want to access (Running status) and click **Connect**:

Instances (1/80) Info Last updated 29 minutes ago Connect Instance state Actions Launch Instances

Find Instance by attribute or tag (case-sensitive)

group-01 Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability zone
☒ ec2-group-01-node-1	i-023c81a1c1fc091ad	Running	t4g.medium	3/3 checks passed	View alarms +	eu-west-3a
☐ ec2-group-01-node-2	i-04646d59e796c29cd	Stopped	t4g.medium	-	View alarms +	eu-west-3a
☐ ec2-group-01-node-3	i-068bb1c8fade5c79c	Stopped	t4g.medium	-	View alarms +	eu-west-3a

2. Select the SSM Session Manager tab and click Connect.

EC2 Instance Connect SSM Session Manager SSH client EC2 serial console

SSM agent status

Introducing Systems Manager just-in-time node access
Move towards zero standing privileges by requiring operators to request access before remotely connecting to instances. [Learn more](#)

Try for free ×

Ping status
Online

Last ping time
Sun Mar 15 2026 17:34:33 GMT-0300 (Brasilia Standard Time)

Session Manager connection status
Connected

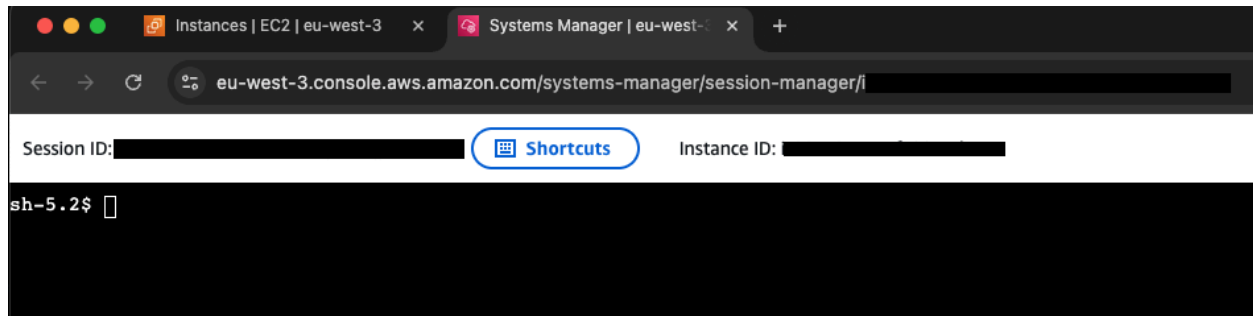
Agent version
3.3.3883.0

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

Cancel Connect

3. A terminal window will appear, and you will be able to access the instance directly through the browser.



Accessing using CLI

Retrieve the instance ID that you want to connect to (i-xxxx) and run this command to open an interactive shell:

...

```
$ aws --profile kubequest2 ssm start-session --target i-xxx
```

...

Step 6: Shared Storage (Amazon EFS)

Each group is provisioned with a shared network storage volume, mounted at `/mnt/efs` on all three nodes (node-1, node-2, node-3). Files saved to this location are accessible from any of your nodes and persist independently of each node's local disk, including across reboots.

```
sh-5.2$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0    4.0M   0% /dev
tmpfs           1.9G   0    1.9G   0% /dev/shm
tmpfs           767M  444K   766M   1% /run
efivarfs        128K   2.6K   126K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1   30G   2.3G   28G    8% /
tmpfs           1.9G   0    1.9G   0% /tmp
/dev/nvme0n1p128 10M   1.4M   8.6M  14% /boot/efi
127.0.0.1:/      8.0E   0    8.0E   0% /mnt/efs
sh-5.2$
```

Storage limit: 20 GB per group

This is a hard limit. You are responsible for monitoring your usage and keeping it under this threshold. Exceeding it will trigger an automated warning notification to the group. Continuing to exceed the limit will result in the storage becoming read-only, at which point no further data can

be written until usage is reduced. Existing files remain readable, and files can still be deleted to free up space and restore write access.

Plan your usage accordingly. Store only what is necessary for your coursework, and remove files you no longer need.

Permissions

If you receive a "Permission denied" error when writing to **/mnt/efs**, run the following command once, from any single node in your group:

```
...
```

```
$ sudo chmod 1777 /mnt/efs
```

```
...
```

This is a one-time action. It does not need to be repeated on the other nodes, and it will persist across reboots.

Notes

- If a write fails unexpectedly, check your current usage against the 20 GB limit before assuming it is a permissions issue.

Appendices

Shutdown script

Every day at 9:00 PM (Paris time), a script shuts down all the instances. We ask you to turn off the instances when you are no longer using it, but this script will do it in case you miss it.

Security (Open Ports)

Your EC2 instances are protected by a strict AWS Security Group that acts as a firewall.

By default, the following ports are open for inbound traffic from the internet to your cluster:

- **Port 80 (HTTP):** Used for web services deployed without SSL/TLS.
- **Port 443 (HTTPS):** Used for secure web services.

All outbound traffic from your instances to the internet is completely allowed. Additionally, all internal network traffic between the 3 instances is fully open, ensuring your Kubernetes deployment can communicate internally without issues.

Creating and Restoring Snapshots

A snapshot is a backup of your instance's disk (EBS volume). If you break your Kubernetes setup or want to save a known-good state before making risky changes, snapshots are your rollback strategy.

Creating a Snapshot

1. Go to EC2 > Snapshots and Create snapshot

In the AWS Console, go to EC2 → Snapshots. Click on the Create snapshot button.

EC2

Dashboard
AWS Global View
Events

▼ Instances

Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

▼ Images

AMIs
AMI Catalog

▼ Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

▼ Network & Security

Snapshots (50)

Snapshot scope

Owned by me

Search

1

Recycle Bin

Actions

Create snapshot

Last updated

1 minute ago

☐	Name	Snapshot ID	Full snapshot size	Volume size	Description
☐	kubequest-21-...	snap-05c797d3f7d20a3b	8.22 GiB	40 GiB	kubequest-21 vm2 monit
☐		snap-012f9f02f46473003	7.13 GiB	40 GiB	node 4 gitops
☐		snap-05e0b159fda2b7339	5.18 GiB	40 GiB	node 3 gitops
☐		snap-089afa1c5e3e7dc06	5.25 GiB	40 GiB	node 2 gitops
☐		snap-07411c1a11c85199f	2.84 GiB	40 GiB	preparation node 3
☐		snap-0b58709f2d539b062	2.85 GiB	40 GiB	preparation node 4
☐		snap-0cc2c0d16433e0377	2.83 GiB	40 GiB	Snapshot vierge node 2
☐	kubequest-21-...	snap-07450a9a685257e68	8.11 GiB	40 GiB	kubequest-21 vm2 monit
☐	kubequest-21-...	snap-01491c68d966326f2	8.11 GiB	40 GiB	kubequest-21 vm2 monit
☐		snap-0263396eef6f47d86	3.96 GiB	40 GiB	cluster ok node 2
☐		snap-0019b9dada68c943a	3.94 GiB	40 GiB	cluster ok node 3
☐		snap-0707780dc3f402110	3.89 GiB	40 GiB	cluster ok node 4
☐		snap-098f6b757ee621d7a	2.87 GiB	40 GiB	preparation node 2

Select a snapshot above.

Select Instance as **Resource Type** and select the instance ID of the instance that you want to take the snapshot. At **Snapshot details**, add a description for the snapshot. This will help you to identify later.

EC2 > Snapshots > Create snapshot

Create snapshot

Source

Resource type

Volume

Create a snapshot from a specific volume.

Instance

Create multi-volume snapshots from an instance.

Instance ID

The instance from which to create multi-volume snapshots.

i-0b0a22c2a90feab72 (ec2-group-41-node-1)

euw1-az1 (eu-west-1)

Snapshot details

Description

Add a description for your snapshot.

255 characters maximum

In **Volumes - optional**, check the **Copy tags** option.

Volumes - optional

By default, all volumes attached to the instance are included in the multi-volume snapshot set. You can optionally exclude the root volume or specific data volumes. You can also indicate whether to copy the tags from the source volumes to the snapshots.

Exclude volumes

Indicate whether to exclude the root volume or specific data volumes from the snapshot set.

Exclude root volume (vol-0ce51ceef7bbcf6a9, Encrypted)

Exclude specific data volumes

Copy tags from source volume

Indicate whether to copy the tags from the source volume to the snapshot.

Copy tags

Then, click on **Create Snapshot** at the bottom of the page.

2. Verify the snapshot created

In the left panel, go to EC2 → Elastic Block Store → Snapshots. Your snapshot will show with a status of pending, then completed once it's ready (usually takes 1-2 minutes).

Snapshots (1) Info Last updated 1 minute ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

Snapshot scope: Owned by me

☐	Name	Snapshot ID	Full s...	Volume size	Description	Storage tier	Snapshot status	Started
☐	Test	snap-090ea1e96226c24a5	-	40 GiB	Test	Standard	Pending	2026/03

Note: If you are struggling in finding the snapshot that you created, go to the instance and copy the Volume ID. Then, go to EC2 → Elastic Block Store → Snapshots, and filter by "Volume ID = <volume_id>"

Snapshots (1) Info Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create snapshot](#)

Snapshot scope: Owned by me

Volume ID = vol-0c370c56eb8ff1d7e [Clear filters](#)

☐	Name	Snapshot ID	Full s...	Volume size	Description	Storage tier	Snapshot status	Started
☐	Test	snap-090ea1e96226c24a5	2.55 GiB	40 GiB	Test	Standard	Completed	2026/03

Restoring the root volume from a Snapshot

- Go to EC2 → Instances. Filter by your group name (e.g., group-01).
- Select the instance you want to restore. Click Actions → Maintain & troubleshoot → Replace root volume.

Instances (1/4) Info Connect Instance state Actions [Launch instances](#)

All states

[Clear filters](#)

☐	Name	Instance ID	Instance state	Instance type	Size (GB)
☒	ec2-group-21-node-1	i-045cf1b38317e1167	Running	t4g.medium	2
☐	ec2-group-21-node-4	i-028c746f5b0344970	Running	t4g.medium	2
☐	ec2-group-21-node-3	i-093fe8592326bb4c0	Running	t4g.medium	2
☐	ec2-group-21-node-2	i-04f65650cf37b7ba5	Running	t4g.medium	2

i-045cf1b38317e1167 (ec2-group-21-node-1)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#)

Instance summary Info

Instance ID: i-045cf1b38317e1167

Public IPv4 address: 13.36.150.93 | [open address](#)

IPv6 address: [Public DNS](#)

Monitor and troubleshoot

- Monitor
 - Manage detailed monitoring
 - Manage CloudWatch alarms
 - Configure CloudWatch agent
- Troubleshoot
 - Get system log
 - Get instance screenshot
 - Connect to serial port
 - Replace root volume**
 - Visit SSM Fleet Manager

- In the dialog:
 - Select Snapshot as the restore source.

- Choose the snapshot you want to restore from (use the snapshot ID or the description you wrote earlier).
- Check the Delete replaced root volume.
- Click **Replace root volume**.

Root volume details

Instance ID
i-045cf1b38317e1167 (ec2-group-21-node-1)

Root volume ID
vol-00824df0726c21012

Recent root volume replacement tasks
▶ replacevol-0815209b9cec7eab6 Successful

Restore
Specify whether to restore the instance's root volume to its initial launch state, from a specific snapshot, or from an AMI.

☐ Launch state
☒ Snapshot
☐ Image

Snapshot
Specify a snapshot to restore the root volume to that state.

Q snap-090ea1e96226c24a5 X

Volume initialization rate (MiB/s) - new, optional | [info](#)
Specify the rate at which the snapshot blocks are to be downloaded from Amazon S3 to the volume. [Additional costs apply](#)

Enter a value

Min: 100 MiB/s, max: 300 MiB/s.

☒ **Delete replaced root volume**
Delete the original root volume (vol-00824df0726c21012) after it has been successfully replaced.

Tags - optional
No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Replace root volume](#)

4. AWS will automatically stop the instance, replace the root volume, and restart it. You can monitor the progress:
- Go to EC2 → Instances → select your instance.
 - Click the Storage tab.
 - **Under Replace root volume tasks** (scroll down entirely), you'll see the status (e.g., in-progress, succeeded).

Instances (1/4) Info Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

group-21 Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availab
☒ ec2-group-21-node-1	i-045cf1b38317e1167	Running	t4g.medium	3/3 checks passec	View alarms +	eu-west
☐ ec2-group-21-node-4	i-028c746f5b0344970	Running	t4g.medium	3/3 checks passec	View alarms +	eu-west
☐ ec2-group-21-node-3	i-093fe8592326bb4c0	Running	t4g.medium	3/3 checks passec	View alarms +	eu-west
☐ ec2-group-21-node-2	i-04f65650cf37b7ba5	Running	t4g.medium	3/3 checks passec	View alarms +	eu-west

i-045cf1b38317e1167 (ec2-group-21-node-1)

U4:3U U5:UU U4:3U U5:UU U4:3U U5:UU

vol-00824df0726c21012 vol-00824df0726c21012 vol-00824df0726c21012

Recent root volume replacement tasks

Filter tasks Replace root volume

Task ID	Task state	Start time	Completion time	Tags
replacevol-0815209b9cec7eab6	Successful	2026-03-20T05:11:20Z	2026-03-20T05:12:01Z	-

- Once the task shows **Successful**, your instance will be back up with the restored volume. Connect via SSH or Session Manager to verify.

Note: The restore process usually takes 2-5 minutes. The original root volume is automatically deleted and replaced by the one created from your snapshot.

Restoring the root volume from its initial state

When replacing the root volume, AWS offers an option to restore from the initial launch state rather than selecting a specific snapshot.

The behavior depends on whether the instance has had its root volume replaced before:

- First time replacing: AWS restores the root volume directly from the original AMI used to launch the instance, effectively reverting it to a clean state.
- Previously replaced: AWS restores from the snapshot that was automatically taken during the last root volume replacement, not the original AMI.

Note: Each time you perform a root volume replacement, AWS automatically captures a snapshot of the current root volume before replacing it. This snapshot becomes the new "initial state" reference for any future restore operations.

Replace root volume [Info](#)

Replace the root volume for an instance with a new (replacement) root volume that is restored to the initial launch state, from a specific snapshot, or from an AMI. Data stored on instance store volumes, and networking and IAM configuration is retained.

Root volume details

Instance ID

 i-0b0a22c2a90feab72 (ec2-group-41-node-1)

Root volume ID

 vol-0ce51ceef7bbcf6a9

Recent root volume replacement tasks

▶ replacevol-06b340d952e5d752a  Successful

Restore

Specify whether to restore the instance's root volume to its initial launch state, from a specific snapshot, or from an AMI.

☒ Launch state

☐ Snapshot

☐ Image

Volume initialization rate (MiB/s) - *new, optional* [Info](#)

Specify the rate at which the snapshot blocks are to be downloaded from Amazon S3 to the volume. [Additional costs apply](#) 

Min: 100 MiB/s, max: 300 MiB/s.

☐ Delete replaced root volume

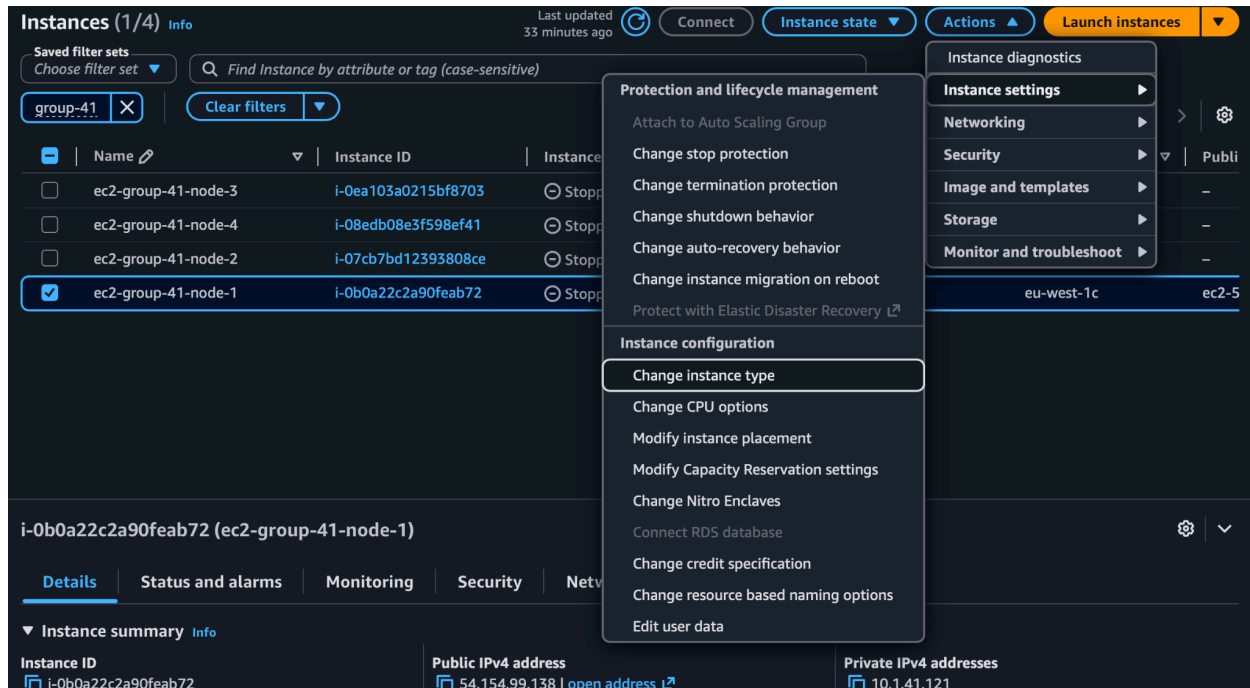
Delete the original root volume (vol-0ce51ceef7bbcf6a9) after it has been successfully replaced.

Changing Instance Type

Due to insufficient capacity errors, we added the ability to change the instance type to c6g.medium or c7g.medium as alternatives to the default t4g.medium. These instance types have fewer resources (1 vCPU and 2 GiB RAM) but are more likely to have available capacity when t4g.medium is unavailable.

Note: Only change the instance type if you are experiencing an insufficient capacity error. The instance must be stopped before proceeding.

1) In the EC2 console, select the instance and go to Actions → Instance Settings → Change instance type.



Instances (1/4) Info

Last updated 33 minutes ago

Find Instance by attribute or tag (case-sensitive)

group-41

Clear filters

	Name	Instance ID	Instance state
☐	ec2-group-41-node-3	i-0ea103a0215bf8703	Stopped
☐	ec2-group-41-node-4	i-08edb08e3f598ef41	Stopped
☐	ec2-group-41-node-2	i-07cb7bd12393808ce	Stopped
☒	ec2-group-41-node-1	i-0b0a22c2a90feab72	Stopped

i-0b0a22c2a90feab72 (ec2-group-41-node-1)

Details | Status and alarms | Monitoring | Security | Networking

Instance summary Info

Instance ID: i-0b0a22c2a90feab72

Public IPv4 address: 54.154.99.138 | open address

Private IPv4 addresses: 10.1.41.121

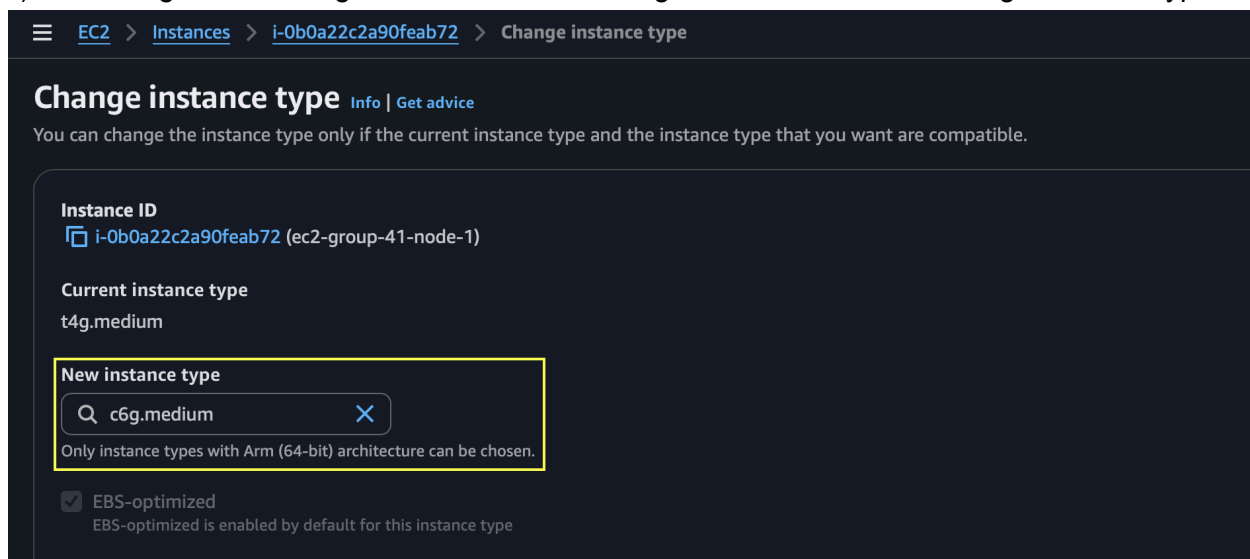
Protection and lifecycle management

- Attach to Auto Scaling Group
- Change stop protection
- Change termination protection
- Change shutdown behavior
- Change auto-recovery behavior
- Change instance migration on reboot
- Protect with Elastic Disaster Recovery

Instance configuration

- Change instance type**
- Change CPU options
- Modify instance placement
- Modify Capacity Reservation settings
- Change Nitro Enclaves
- Connect RDS database
- Change credit specification
- Change resource based naming options
- Edit user data

2) Select c6g.medium, c7g.medium, or revert to t4g.medium, then click Change instance type.



Change instance type Info | Get advice

You can change the instance type only if the current instance type and the instance type that you want are compatible.

Instance ID

i-0b0a22c2a90feab72 (ec2-group-41-node-1)

Current instance type

t4g.medium

New instance type

c6g.medium

Only instance types with Arm (64-bit) architecture can be chosen.

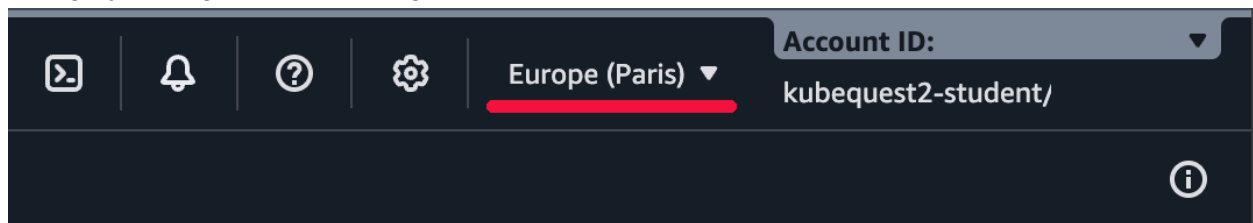
☒ EBS-optimized
EBS-optimized is enabled by default for this instance type

3) Start the instance normally. It will now run on the selected instance type.

Recurring issues

I don't see my instances in AWS

If you don't see your resources in AWS, it's probably because you are in the wrong AWS Region. Always make sure to be in the region your group was assigned to. You can see and change your region in the top right corner of the console.



I receive an “insufficient capacity” error when starting my EC2

AWS has limited capacity in each region for each instance type. If your AWS instance type region pool is exhausted, you can either retry a few seconds/minutes later or follow instructions from the “Changing Instance Type” section.